SHROTI TELECOM · PRODUCT DATASHEET

TPMS

THEFT PREVENTION MANAGEMENT SYSTEM

Real-Time Safety, Anytime, Anywhere

An intelligent, industrial-grade security platform engineered to protect telecom tower assets from theft, intrusion, and infrastructure loss — with 24×7 SOC-backed remote monitoring and multi-channel alerting.

SHROTITELECOM

TELECOM & IOT INFRASTRUCTURE · EST. 2012

TPMS V7.0 | INDUSTRIAL GRADE | SOC ENABLED

01 PRODUCT OVERVIEW

TPMS (Theft Prevention Management System) is Shroti Telecom's flagship security solution purpose-built for telecom tower sites, remote infrastructure, and industrial deployments. It integrates a rugged hardware controller with cloud-connected software and a dedicated 24×7 Security Operations Center (SOC) to deliver end-to-end asset protection. The system monitors for battery theft, diesel fuel pilferage, cable tampering, equipment removal, and unauthorized intrusion in real time — triggering multi-channel alerts instantly upon detection. Engineered for the harshest field conditions, TPMS operates autonomously with internal battery backup and requires minimal on-site intervention.

- 🔋 Battery Theft Prevention
- ⚡ Cable & Copper Theft
- ⛽ DG / Diesel Theft
- 📡 Radio Equipment Theft
- 🚪 Intrusion Detection
- ☀️ Solar Panel Protection
- 🏗️ Infrastructure Theft

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02 KEY FEATURES

📡
REAL-TIME MONITORING
Continuous site surveillance with instant event detection and response.

🔌
16 ALARM INPUTS
Multi-sensor support: door, motion, fire, tamper, battery & power.

🔗
MODBUS RS485
Industry-standard serial communication for reliable integration.

💬
MULTI-CHANNEL ALERTS
WhatsApp, SMS & Robo Voice Call notifications on alarm events.

🔔
BUILT-IN HOOTER
125 dB high-power on-site alarm for active theft deterrence.

🔋
BATTERY BACKUP
1200 mAh internal battery with BMS — up to 10 hours backup time.

☁️
CLOUD & API READY
Seamless cloud integration with REST API support for enterprise platforms.

🌡️
HARSH ENVIRONMENT
Rated −10°C to +55°C. Designed for outdoor & remote deployments.

03 SYSTEM ARCHITECTURE

SENSORS
Door · Motion
Fire · Tamper
Battery · Power

→

TPMS CONTROLLER
V7.0 MCU RL78
16-Input Processing
RS485 MODBUS

→

CLOUD PLATFORM
API Gateway
Event Engine
Data Analytics

→

SOC CENTER
24×7×365
Alarm Verification
Incident Response

→

ALERTS & APP
WhatsApp · SMS
Robo Call
Mobile App

Field-deployed sensors feed alarm data into the TPMS V7.0 controller via wired inputs. The controller processes events locally — triggering the 125 dB hooter — and simultaneously transmits structured data to the Shroti Cloud over cellular/RS485. The cloud platform verifies events and routes them to the SOC Command Centre, which dispatches multi-channel alerts (WhatsApp, SMS, Robo Call, App notification) and escalates field response within defined SLA windows.

04 TECHNICAL SPECIFICATIONS

PARAMETER	SPECIFICATION
SYSTEM OVERVIEW	
Model	TPMS V7.0
Processor	16-bit MCU – Renesas RL78
Input Voltage	48V DC
Operating Temperature	−10°C to +55°C
PHYSICAL SPECIFICATIONS	
Dimensions (L × H × W)	22 cm × 11 cm × 16 cm
Weight	1.57 kg
Antenna	Internal & External support
POWER & BACKUP	
Internal Battery	1200 mAh with Battery Management System (BMS)
Backup Duration	Up to 10 hours (under typical load)
ALARM & MONITORING	
Alarm Inputs	16 dynamic alarm input ports
Supported Sensors	Door, Motion, Fire, Tamper, Battery monitoring, Power monitoring
Local Alarm	Built-in Hooter — 125 dB high-power
COMMUNICATION & CONNECTIVITY	
Communication Protocol	MODBUS – RS485

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PARAMETER	SPECIFICATION
Alert Channels	Robo Call, WhatsApp, SMS, Mobile App notifications
Cloud Integration	REST API support; cloud-hosted dashboard
INDICATORS	
LED Indicators	Power Status, Network Status, Alarm Status

05 APPLICATIONS

Telecom BTS & Tower Sites
Battery, DG, cable, and RRU equipment protection for 2G/4G/5G sites.

Battery Swapping Stations
Real-time monitoring of high-value Li-ion and VRLA battery banks.

Solar Energy Sites
Dual-logic solar panel protection using NO/NC resistance wire sensing.

Remote Infrastructure
Unmanned shelter, outdoor enclosure & cabinet security in remote areas.

Industrial Security
Perimeter protection, fire detection & tamper monitoring for industrial assets.

IoT & Smart Sites
Integration into broader IoT automation frameworks via API and MODBUS.

06 BUSINESS BENEFITS

Theft Reduction
Deter and detect asset theft before losses occur.

Cost Savings
Significant reduction in replacement & downtime costs.

Remote Visibility
Full site awareness from any location via dashboard & app.

Faster Response
Immediate multi-channel alerts minimize response lag.

24×7 SOC
Round-the-clock monitoring by trained security analysts.

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07 SECURITY OPERATIONS CENTER (SOC)

SOC CAPABILITIES

- 24×7×365 centralised monitoring of all field sites
- Real-time detection of alarms: theft, fire, and communication loss
- Automated WhatsApp, SMS & Voice Call notifications
- Escalation to field teams within defined SLA timelines
- Incident verification, ticketing, and closure tracking
- Proactive monitoring using historical telemetry trends
- Remote dashboard access with admin-level control

SOC PERFORMANCE

24/7
CONTINUOUS MONITORING

365
DAYS PER YEAR

Instant
ALERT ACKNOWLEDGEMENT

SLA
ESCALATION PROTOCOL

ALERTS & NOTIFICATIONS

WhatsApp Alerts
Instant message notification to registered contacts with event details and timestamps.

SMS Alerts
Carrier-independent SMS dispatch ensuring delivery even in low-data environments.

Robo Voice Call
Automated voice calls to on-call personnel for high-priority alarm escalation.

Mobile App
Push notifications and live event feed via the Shroti Telecom admin application.

09 COMPLIANCE & RELIABILITY

Industrial-Grade Design — Built to withstand demanding telecom field environments with a rugged enclosure and robust internal components.

Stable Communication — RS485 MODBUS protocol ensures long-distance, noise-resistant data transmission in electrically harsh environments.

Weather-Resistant — Designed for outdoor and semi-outdoor deployment, tolerating temperature extremes, dust, and humidity.

Battery Management System (BMS) — Intelligent BMS protects the internal backup battery from over-charge, over-discharge, and thermal events.

Lightning Protection — Infrared sensors feature built-in lightning protection for installations in exposed locations.

Standalone Operation — TPMS operates as a fully autonomous unit; no external controller required for core protection functions.

10 COMPANY PROFILE

SHROTITELECOM

Established 2012 · Bhopal, Madhya Pradesh

13+ Years in telecom infrastructure

300+ Trained PAN-India field engineers

24/7 SOC Command Centre

PAN India operations & O&M capability

Established in 2012, Shroti Telecom began as a passive infrastructure solutions provider with a vision to deliver rugged, easy-to-integrate, and cost-effective products for telecom tower operations. Over 13+ years, it has transformed into a full-stack telecom infrastructure and security solutions company with several first-in-India innovations.

The company operates an integrated manufacturing and repair division ensuring quality control and fast turnaround, supported by a 24×7 SOC Command Centre for real-time monitoring and incident response across all deployed sites.

- Theft Prevention & Security
- Remote Monitoring (RMS)
- IoT Automation
- Telecom Shelter Manufacturing
- Smart Power Electronics
- Field O&M Services

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